

Wave forcing $M(t)$
Damped roll / pitch modes

Ray acquisition t_i, α_i, ρ_i
Pose R_i , origin s_i

Map correction
+ coverage / ambiguity flag

Midpoint boresight applied to whole scan

Lever arm ℓ

Elevation changes
before mapping

Ray at t_i

Known plane $z = 0$

Vessel / IMU

